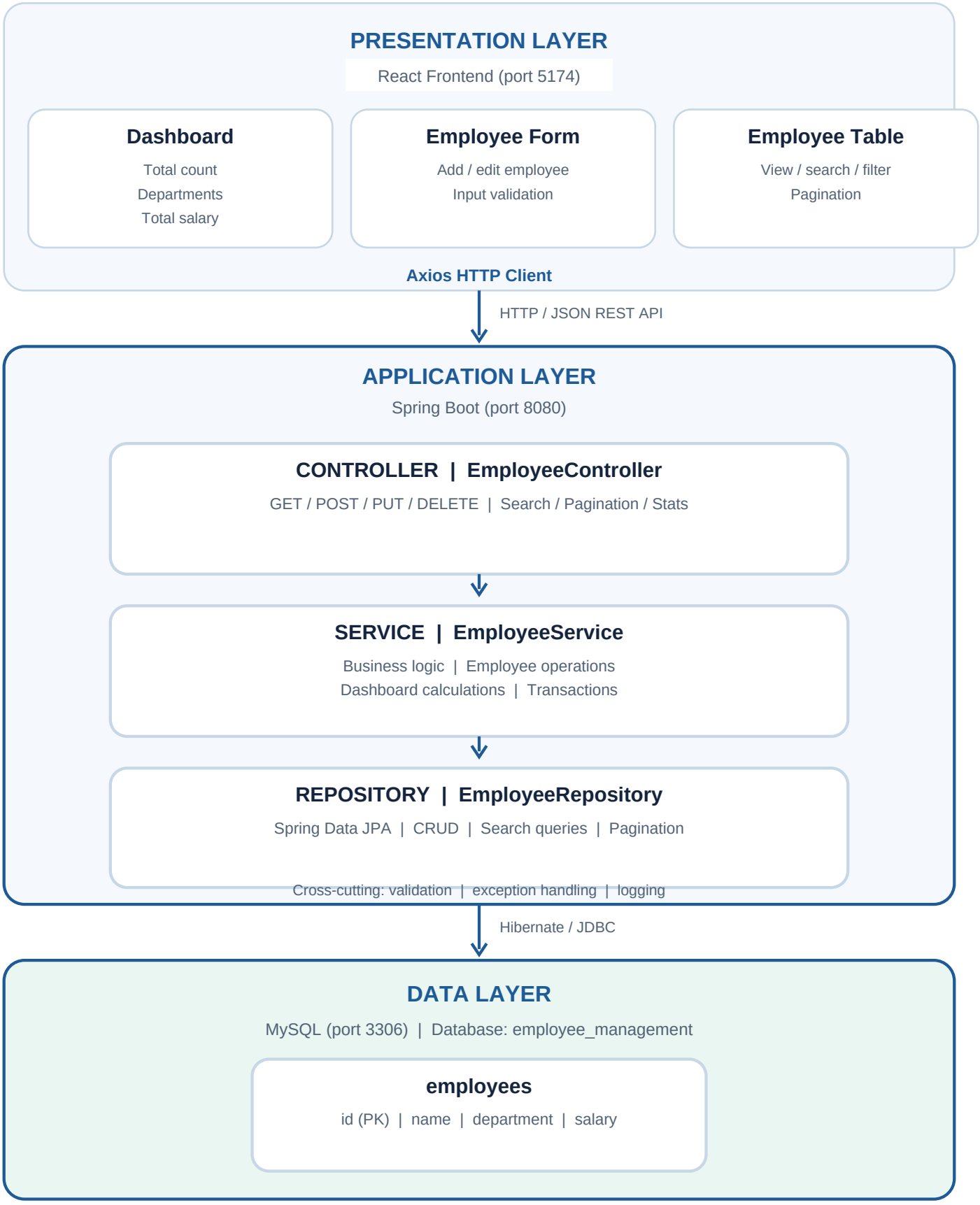
1. System Architecture

Three-tier architecture | React + Spring Boot + MySQL



2. Add Employee - Data Flow

Request and response lifecycle | HTTP 201 Created

1. USER / REACT FRONTEND

Click Add Employee; enter name, department, salary
React validates form input

2. AXIOS HTTP CLIENT

POST /api/employees
Send employee details as JSON

3. EMPLOYEE CONTROLLER

Receive request body as EmployeeRequest
@Valid checks request fields

4. EMPLOYEE SERVICE

Create Employee entity from request
Call repository.save(employee)

5. EMPLOYEE REPOSITORY

Spring Data JPA / Hibernate
Persist entity using INSERT

6. MYSQL DATABASE

Store employee record
Generate employee ID

7. SPRING BOOT RESPONSE

Return saved employee as JSON
HTTP 201 Created

8. REACT FRONTEND

Receive response and refresh employee list
Display updated dashboard

Example input: Rabindra | IT | 45000